

PalettePilot

A Real-Time Spatial AI Painting Copilot

Turn any reference image into a personalized painting lesson that sees your progress, decides the next best step, and projects guidance directly onto your physical workspace.

PROBLEM

- A reference image shows the final artwork, but not the process required to paint it.
- Painters struggle with what to paint first, color mixing, brush choice, blending, stroke direction, layering, values, and technique.
- Video tutorials are fixed and usually exist only for a specific painting; they cannot see or adapt to the painter's actual progress.

SOLUTION

- Upload any painting, artwork, or photograph and select the painting medium.
- AI decomposes the reference into a practical painting strategy: composition, layers, palette, brush/technique, stroke direction, and sequence.
- As the user paints, Sprout observes the real work. AI evaluates progress and generates the next best action instead of following a rigid tutorial.
- Sprout projects spatial guidance onto the table/canvas area while ElevenLabs provides hands-free voice coaching.

Core Experience: Closed-Loop Physical Tutoring

Reference Image	AI Plans	Sprout Guides	User Paints	Camera Observes	AI Adapts	Next Step
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What the User Sees

ON THE WORKSPACE	ON THE SCREEN / TABLE	BY VOICE
Projected composition guides, next paint region, blend zones, stroke direction, approximate boundaries, and correction overlays.	Next step, target color, suggested color ratios, brush type, technique, pressure, and short explanation of why the step matters.	Hands-free prompts such as "What's next?", "Show me where", "Is this color right?", and "Why doesn't mine look like the reference?"

Technology Roles

HP Sprout	Gemini	ElevenLabs	Backboard
Camera + spatial projection onto the physical workspace.	Multimodal analysis of reference + current painting; planning and real-time critique.	Natural hands-free voice instructor while the user's hands are occupied with paint.	Persistent artist memory: skill level, recurring mistakes, mastered techniques, preferences and prior sessions.

Initial POC for the Hackathon

Scope: Acrylic landscape painting, one reference image at a time, with a 5-step adaptive workflow.

- Upload a landscape reference. Gemini returns a structured 5-step painting plan with regions, colors, brush/technique and success criteria.
- Sprout projects Step 1 guidance and the workspace displays the target color/technique. ElevenLabs narrates the instruction.
- User paints, then says/clicks "Check my progress." Sprout captures the current work.
- Gemini compares the reference, current painting and current step; it returns READY or one highest-impact ADJUSTMENT.
- Sprout highlights the correction region. Once corrected, the system advances and stores learning observations in Backboard.

DIFFERENTIATOR: YouTube teaches a predetermined painting. Generative AI creates the artwork for you. PalettePilot watches you create and teaches you what to do next.